

$$X = ((1, 3), (4, 10))$$

$$y = (5, 6)$$

$$m \in [-1, 2]$$

$$b \in [1]$$

$$f(m, m_2, b) = m_2 x_2 + x_2 m_2 + b$$

$$\hat{y} = mX + b, \quad J(m, b) = \frac{1}{n} \sum_i (\hat{y}_i - y_i)^2$$

$$\text{Let } e = \hat{y} - y$$

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \times \frac{\partial \hat{y}_i}{\partial m}; \quad \frac{\partial \hat{y}}{\partial m} = \frac{\partial (mX + b)}{\partial m} = X \Rightarrow \frac{\partial \hat{y}_i}{\partial m} = X_i$$

$$= \frac{2}{n} \sum_i e_i X_i$$

$$= \frac{2}{n} \sum e X = \frac{2}{n} X^T e$$

$$\frac{\partial J}{\partial b} = \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \frac{\partial \hat{y}_i}{\partial b}; \quad \frac{\partial \hat{y}}{\partial b} = \frac{\partial (mX + b)}{\partial b} = 1$$

$$= \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \times 1$$

$$= \frac{2}{n} \sum e$$

$$X = ((1, 3), (4, 10))$$

$$y = (5, 6)$$

$$\text{Let } \alpha = 0.01$$

$$e = \hat{y} - y$$

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e$$

$$\frac{\partial J}{\partial b} = \frac{2}{n} \sum e$$

$$m_{\text{new}} = m_{\text{old}} - \alpha \frac{\partial J}{\partial m} = m_{\text{old}} - \frac{2\alpha}{n} X^T e$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{\partial J}{\partial b} = b_{\text{old}} - \frac{2\alpha}{n} \sum e$$

Iteration 1

Iteration 1

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ = \begin{bmatrix} -1 + 6 + 1 \\ -4 + 20 + 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$e = \hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\left. \begin{aligned} x^T e &= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 34 \\ 113 \end{bmatrix} \\ \sum e &= 12 \end{aligned} \right\} \begin{aligned} m &= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 34 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.34 \\ 0.87 \end{bmatrix} \\ b &= 1 - 0.01 \times 12 = 0.98 \end{aligned}$$

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$$X = \{(1, 3), (4, 10)\}$$

$$Y = \{(5, 6)\}$$

$$m = [1, 2]$$

$$b = 1$$

$$\text{Let } e = \hat{y} - y$$

$$\frac{dJ}{db} = \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \frac{d\hat{y}_i}{db} \quad ; \quad \frac{d\hat{y}}{db} = \frac{d(mx + b)}{db} = 1$$

$$= \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \times 1$$

$$= \frac{2}{n} \sum e$$

$$X = \{(1, 3), (4, 10)\}$$

$$Y = \{(5, 6)\}$$

$$\text{Let } \alpha = 0.01$$

$$e = \hat{y} - y$$

$$\frac{dJ}{dm} = \frac{2}{n} X^T e$$

$$\frac{dJ}{db} = \frac{2}{n} \sum e$$

$$J(m, b) = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\frac{dJ}{dm} = \frac{1}{n} \sum_i 2(\hat{y}_i - y_i) \times \frac{dy_i}{dm} \quad ; \quad \frac{dy_i}{dm} = x_i$$

$$= \frac{2}{n} \sum e x_i$$

$$= \frac{2}{n} \sum e x = \frac{2}{n} X^T e$$

$$m_{\text{new}} = m_{\text{old}} - \alpha \frac{dJ}{dm} = m_{\text{old}} - \frac{2\alpha}{n} X^T e$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{dJ}{db} = b_{\text{old}} - \frac{2\alpha}{n} \sum e$$

Iteration 2

$$m_{\text{new}} = [-1.45, 0.87]$$

$$Y = \{(5, 6)\}$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$b = 0.88$$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} m + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \Rightarrow \hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} = \begin{bmatrix} -9.45 + 2.61 + 0.88 \\ -5.8 + 8.7 + 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

$$e = \hat{y} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$X^T e = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$\sum e = -5.18$$

$$\Rightarrow m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -1.3316 \\ 0.9318 \end{bmatrix}$$

$$b = 0.88 - 0.01(5.18) = 0.9318$$

$$m = \begin{bmatrix} -1.3316 \\ 0.9318 \end{bmatrix}$$

$$b = 0.9318$$

iteration 3:

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} m + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$m = \begin{bmatrix} -2.3316 \\ 1.1808 \end{bmatrix} \Rightarrow \hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} -1.3316 + 3.5424 + 0.9318 \\ -5.3284 + 11.808 + 0.9318 \end{bmatrix}$$
$$= \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

$$e = \hat{y} - y = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$X^T e = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix} \Rightarrow m = \begin{bmatrix} -2.3316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix} = \begin{bmatrix} -2.369562 \\ 1.095182 \end{bmatrix}$$
$$\sum e = -0.444 \quad b = 0.9318 - 0.01 \bar{x}(-0.444) = 0.93624$$